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Moving Border Ownership for Event-based Motion Segmentation

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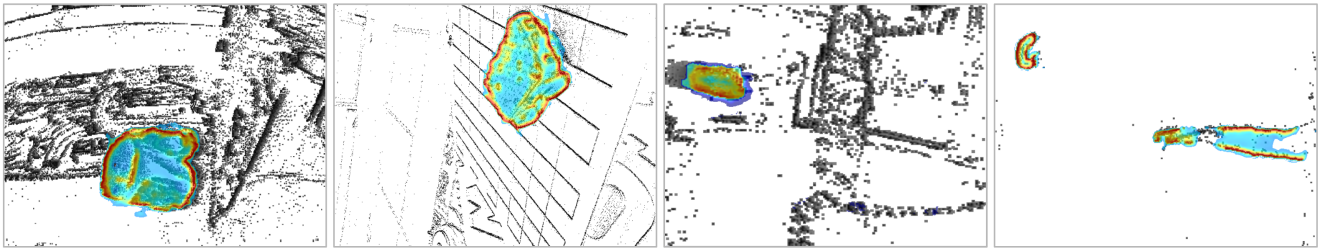


Figure 1. Moving border ownership detection on real event-based datasets. Our model, trained solely on synthetic data, generalizes zero-shot to EVIMO1, EVIMO2, EED, and EMSMC datasets. Red boundaries indicate moving object edges; yellow-to-blue gradients visualize the predicted ownership field, showing which side of each border belongs to the moving foreground object.

Abstract

Event cameras provide accurate information at motion boundaries—exactly where disentangling ego-motion, object motion, and border ownership determines segmentation quality. We argue that the missing ingredient in dynamic scene interpretation is moving border ownership: detecting motion boundaries and assigning which side is foreground so occlusions are resolved by design. Traditional geometric motion segmentation pipelines (e.g., flow clustering, simple motion models) remain assumption-heavy and slow, while deep models often fail to generalize across sensors or datasets. We introduce a lightweight, ownership-aware predictor trained solely on synthetic events with perfect supervision for boundaries, ownership, and motion, generated via a Blender pipeline. Its key targets—a signed-distance ownership field and a motion mask—focus learning where events occur and yield stable gradients. The model runs in real time and generalizes without tuning: trained on synthetic events, it achieves zero-shot transfer on EED, EVIMO1, EVIMO2, and EMSMC, delivering state-of-the-art performance. By casting motion segmentation as ownership-aware edge understanding, we combine the robustness of model-based reasoning with the scalability of learning.

1. Introduction

Neuromorphic event-based sensors have attracted growing attention in the computer vision and robotics communities, driving advances in visual motion analysis applications [5, 10]. Among the fundamental problems is object motion segmentation—detecting and localizing independently moving objects within a scene—and by their very nature, the sparse, temporally precise events produced by these sensors offer a rich and information-efficient representation for this task.

When the camera is static, this task is straightforward: events are primarily generated along the contours and texture of moving objects, making segmentation easier. However, when the sensor itself is in motion, the challenge increases substantially [22]. Then, the event stream captures both the camera's ego-motion and the movement of objects in the environment, which must be disentangled [35].

In contemporary computer vision, moving-object detection is often treated as an object-detection or semantic-segmentation problem, relying mainly on appearance rather than motion cues. Earlier classical methods instead inferred independent motion from geometric constraints, but only under restrictive assumptions about camera or scene dynamics. Event-based vision has largely adopted similar structure-from-motion strategies, using motion clustering and simplified motion models [3, 32, 41], which re-

main assumption-heavy and computationally demanding. More recently, learning-based approaches have been explored [8, 20, 37], though their generalization capabilities are still limited.

Inspired by biological vision, we propose a lightweight and efficient mechanism for detecting moving objects and their boundaries, including border ownership—that is, determining which side of a border belongs to the foreground or background. This capability is crucial for handling complex dynamic interactions, such as when objects move in front of or behind each other. Our approach leverages a MobileNet-based LSTM architecture trained on temporal time slices. A large synthetic dataset with diverse event patterns (scenes, textures, motions, lighting, bias settings) enables our model to learn robust motion and border representations.

Retinal Ganglion Cells: Detecting independently moving objects is a fundamental visual capability. Pioneering neuroscience work by Meister and colleagues showed that the segregation of object and background motion begins as early as the retina [23], where a subset of ganglion cells responds selectively to object motion and enables early foreground–background separation.

These findings suggest that moving-object detection may function as a simple, robust early-stage mechanism supporting higher-level processes such as tracking, and structure-from-motion. Event-based data are well suited for such a mechanism, providing high-temporal-resolution information exactly along motion boundaries. Yet existing learning-based approaches have struggled to generalize broadly. We argue that effective motion segmentation must learn not only moving-object locations but also their boundaries and border ownership, mirroring key computations in biological vision.

Our main contributions are as follows:

- **Moving border-ownership for events.** We formulate motion segmentation as ownership-aware border prediction, jointly estimating motion boundaries and which side belongs to the moving object, which is a cue defined by depth; this yields stronger, more transferable event-only segmentation than prior approaches.
- **Synthetic supervision that transfers.** A Blender pipeline renders photorealistic indoor scenes with perfect border-ownership and motion labels; models trained only on this synthetic data generalize zero-shot across EVIMO1/2, EED, and EMSMC.
- **Real-time streaming with optional few-shot adapters.** A lightweight time-surface + ConvLSTM stream runs in real time, and ConvLoRA enables few-shot adaptation on EVIMO2, improving from strong zero-shot to state-of-the-art event-only performance.

2. Related Work

2.1. Event-Based Perception and Representations

Event cameras produce sparse asynchronous streams of brightness-change events. Event representations are typically either stream-based, operating directly on events, or frame-based, aggregating events into images or voxel grids [10]. Time surfaces encode recent event history for motion analysis [17, 29], while voxel grids preserve finer timing for learning-based flow, depth, and egomotion [42, 43]. We adopt a temporal time-surface representation that preserves fine motion structure while remaining lightweight for real-time processing.

2.2. Border Ownership and Occlusion Reasoning

Assigning which side of a contour "owns" a border is fundamental to objectness and occlusion resolution. Border-ownership signals arise early in visual cortex and support figure–ground organization and motion layer assignment [40]. The concept has been explored for images, for example in [34], where ownership cues were inferred using random forests.

In computer vision, geometry- and learning-based approaches infer occlusion boundaries and their ownership/orientation from images and videos [30, 36]. In event data—where edges dominate and motion is explicit—ownership is rarely modeled explicitly. Prior learning pipelines for moving object detection typically produce only moving-object masks [11, 18, 20, 21, 26]. Our approach elevates border and ownership to first-class representations by learning an ownership-signed border field tightly coupled to motion grouping, allowing occlusion reasoning to emerge naturally as a by-product of segmentation.

In images, border-ownership reflects relative depth along edges and is largely conveyed by static appearance cues such as convexity, contrast, and texture or intensity gradients [24]. In contrast, motion information alone is often insufficient to disambiguate boundaries from textures, particularly when depth differences are subtle. For example, in static scenes where apparent motion arises solely from camera movement [2], event data do not provide enough evidence for reliable ownership inference at all boundaries. The concept introduced here focuses specifically on learning border and ownership representations for moving objects, where motion cues provide unambiguous, physically grounded evidence of figure–ground separation.

2.3. Motion Segmentation with Events

Event-based motion segmentation has progressed from model-based optimization to learned methods. Contrast maximization fits motion models by warping events under candidate motions and selecting parameters that sharpen the integrated image [9]; geometric clustering then extends this

idea to jointly estimate motions and assign events to motion layers [32, 41]. Although these classical/unsupervised pipelines require no training data, they typically depend on scene-specific choices such as layer count, temporal windows, and thresholds. Learned methods instead predict independent-motion masks from event representations, spanning supervised and self-/unsupervised formulations; representative milestones include EV-IMO's dataset and pipeline [20], event-surface-based graph segmentation [21], and spiking models for low-latency inference [26]. More recent work broadens the application scope (e.g., EV-IMO indoors and DSEC-MOTS outdoors) but still trains per dataset rather than demonstrating cross-dataset zero-shot transfer [11]. Unsupervised domain adaptation addresses this shift by transferring from labeled RGB/flow to unlabeled events, with strong gains on EVIMO2 and MOD++ [18], but still requires an adaptation stage. In contrast, our goal is ownership-aware motion segmentation trained purely on synthetic data and evaluated zero-shot across datasets and sensors.

3. Method

We present our approach in a natural data-flow order: we first define the event stream and windowing conventions (Sec. 3.1), describe the input representation (Sec. 3.2), specify the ground-truth labels for supervision (Sec. 3.3), detail the network architecture (Sec. 3.4), introduce LoRA for bridging the synthetic-to-real domain gap (Sec. 8.1), and finally define the training objectives (Sec. 3.6).

3.1. Notation and Task Setup

Event stream. We observe an asynchronous event sequence

$$\mathcal{E} = \{(x_i, y_i, t_i, p_i)\}_{i=1}^{N}, \qquad p_i \in \{-1, +1\},$$

where (x_i, y_i) are pixel coordinates, t_i is the timestamp, and p_i is the polarity.

Temporal windows. We integrate events over fixed-duration windows

$$W_k = [\tau_k,\ \tau_k + \Delta t),$$

and align all labels to the end-of-window timestamp $(\tau_k + \Delta t)$. This convention is applied consistently during preprocessing, training, and evaluation to eliminate start/end misalignment with frame-based ground-truth masks. For datasets with nonuniform annotation times (e.g., EED), we instead construct windows at the intervals between successive ground-truth timestamps, preserving the same end-of-window alignment.

3.2. Input Representation

Time surface. For each window W_k, we construct a single-channel, normalized time surface $T_k \in [0, 1]^{H \times W}$ that encodes the recency of the last event at each pixel u:

$$T_k(u) \ := \ \frac{t_{\text{last}}(u) - \tau_k}{\Delta t},$$

where $t_{\text{last}}(u)$ is the timestamp of the most recent event at pixel u within W_k. Pixels that receive no event during the window are set to zero. Because temporal context across windows is maintained by the recurrent hidden state (Sec. 3.4), no explicit multi-frame stacking is needed, keeping the input a single channel and the preprocessing minimal.

3.3. Label Construction

At each end-of-window timestamp $(\tau_k + \Delta t)$, we derive two dense supervision targets from the available annotations: an ownership-signed distance field for borders and a binary motion mask.

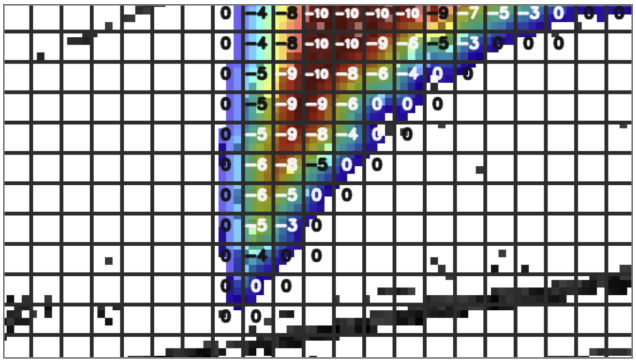


Figure 2. Border ownership label illustration (zoomed in). Grid labels indicate the quantized ground-truth border ownership value for each cell.

Border ownership-signed distance field. Given the instance segmentation and depth maps provided by the synthetic renderer, we assign border ownership (Fig. 2) as follows. A border pixel lies at the boundary between two or more instances in a 4-connected neighborhood. Each such pixel is *owned* by the instance that is closest in depth (to the virtual camera) among its neighbors. Owned-border pixels receive a target value of 0. For every pixel that lies inside a foreground instance, we compute the Euclidean distance to the nearest owned border of that instance via a distance transform, negate it, and clamp the result to $[-10, 0]$. The resulting signed distance field therefore transitions smoothly from 0 at borders to -10 deep inside objects. Background and unlabeled regions (e.g., scene leakage with infinite depth) are assigned a sentinel value and excluded from supervision.

Motion mask. We additionally construct a binary mask $m_k \in \{0, 1\}^{H \times W}$ indicating moving-object regions (1 for foreground motion, 0 for static background). For synthetic data, this mask is provided directly per frame. For

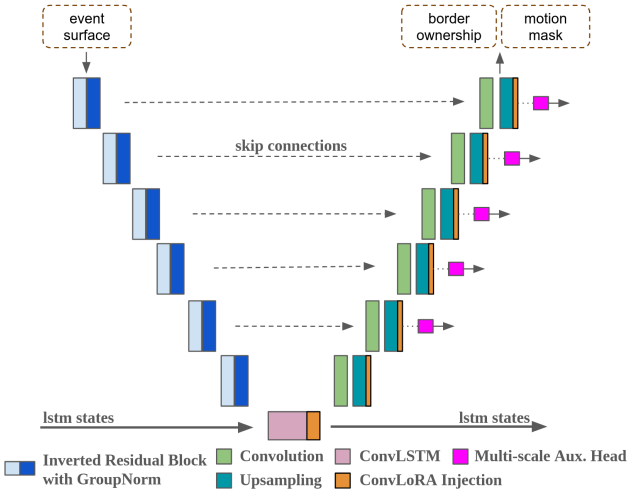


Figure 3. Network architecture. The encoder extracts spatial features from the time surface; the ConvLSTM bottleneck maintains temporal context across windows; the decoder produces border-ownership and motion-mask predictions at full resolution. Auxiliary heads (magenta) provide deep supervision at intermediate scales.

EVIMO2, it is derived on the fly from per-object pose trajectories, using linear and angular velocities over W_k to classify each instance as moving or static.

3.4. Model Architecture

Figure 3 illustrates the overall architecture, which consists of a spatial encoder, a recurrent temporal bottleneck, and a dual-head decoder.

Encoder. We adopt a lightweight MobileNetV3-Large backbone [12] operating on the single-channel time surface T_k. Group Normalization [38] replaces all BatchNorm layers, ensuring stable behavior during streaming inference with batch size of one.

Temporal bottleneck. At the encoder output, a ConvLSTM [28] cell integrates information across successive windows:

$$h_k \;:=\; \mathrm{ConvLSTM}\big(\phi(T_k),\; h_{k-1}\big),$$

where $\phi(\cdot)$ denotes the encoder and h_k is the hidden state at window k. The state rolls forward within a scene and is reset at scene boundaries, providing the model with temporal memory without requiring multi-frame input stacking.

Decoder and prediction heads. A U-Net-style decoder with skip connections progressively upsamples the bottleneck representation to the input resolution and feeds two parallel prediction heads:

$$\hat{b}_k \;:=\; g_b(h_k), \qquad \hat{m}_k \;:=\; \sigma\big(g_m(h_k)\big),$$

where g_b and g_m are convolutional predictors for the border-ownership field and the motion mask, respectively, and σ is the sigmoid function. In addition to these final outputs, auxiliary prediction heads are attached at intermediate decoder stages (shown in magenta in Fig. 3). These heads produce border and motion predictions at their native scales and are supervised with the same loss forms (Sec. 3.6), improving gradient flow and encouraging sharp border structures to emerge early in the decoding process.

3.5. Parameter-Efficient Adaptation with ConvLoRA

For optional few-shot adaptation to real event data, we use convolutional low-rank adapters (ConvLoRA) [1]. After synthetic pretraining, the backbone is frozen and only lightweight adapters are optimized. We insert them in the decoder and prediction heads, where the domain shift is most pronounced, while keeping the generic encoder fixed. This improves EVIMO2 performance with only 1.3% additional trainable parameters. Full formulation and training details are deferred to the supplementary material.

3.6. Training Objectives

Let $\Omega \subset \{1,\dots,H\} \times \{1,\dots,W\}$ denote the set of pixels with valid supervision (sentinels excluded). We optimize a combination of a border regression loss and a binary segmentation loss, applied at multiple scales.

Border-ownership loss. We regress the predicted signed distance field to its ground-truth counterpart (Sec. 3.3) with a spatially weighted mean squared error:

$$\mathcal{L}_b \;:=\; \frac{1}{|\Omega|} \sum_{u \in \Omega} \omega_k(u)\,\big(\hat{b}_k(u) - b_k(u)\big)^2.$$

The weight $\omega_k(u)$ emphasizes a narrow band around owned borders, where precise localization is most critical: pixels within 10 pixels of a border receive a weight of 10, and all remaining pixels receive a weight of 1.

Motion-mask loss. The motion head is supervised with standard pixelwise binary cross-entropy:

$$\mathcal{L}_m \;:=\; -\frac{1}{|\Omega|} \sum_{u \in \Omega} \Big(m_k(u) \log \hat{m}_k(u) + \big(1 - m_k(u)\big) \log\big(1 - \hat{m}_k(u)\big)\Big).$$

Multi-scale aggregation. Let $\mathcal{S}$ index the set of auxiliary decoder outputs. Each auxiliary head produces its own border and motion predictions $\hat{b}_k^{(s)}$, $\hat{m}_k^{(s)}$ at scale s, which are bilinearly upsampled to full resolution, masked by Ω, and supervised with the same loss forms, yielding $\mathcal{L}_b^{(s)}$ and $\mathcal{L}_m^{(s)}$. The total per-window objective is

$$\mathcal{L}_k \;:=\; \mathcal{L}_b + \mathcal{L}_m + \sum_{s \in \mathcal{S}} \alpha_s \big(\mathcal{L}_b^{(s)} + \mathcal{L}_m^{(s)}\big),$$

where $\alpha_s = 0.5$ down-weights the auxiliary losses relative to the main heads. We minimize $\sum_k \mathcal{L}_k$ over all training windows.

4. Synthetic Event Dataset

To train ownership-aware models with perfect supervision, we generate a large-scale synthetic dataset using a

Table 1. Dataset comparison: resolution, sensor type, ground truth availability, and scene characteristics.

Dataset	Resolution	Sensor	GT Mask	GT Pose	GT Depth	Scenes	Objects	Dataset Size
Synthetic (Ours)	640×480	Simulated	✓	✓	✓	18	1000+†	1.6TB
EED [19]	240×180	DAVIS 240B	BoundingBox	×	×	1	4	59MB
EMSMC [32]	346×260	DAVIS 346	×	×	×	2	4	31MB
EVIMO1 [20]	346×260	DAVIS 346C	✓	✓	✓	1	3	19GB
EVIMO2* [4]	640×480	Prophesee Gen3	✓	✓	✓	1	20+	5.1GB

†Randomly sampled from 52472 objects subset *Motion Segmentation Sequences

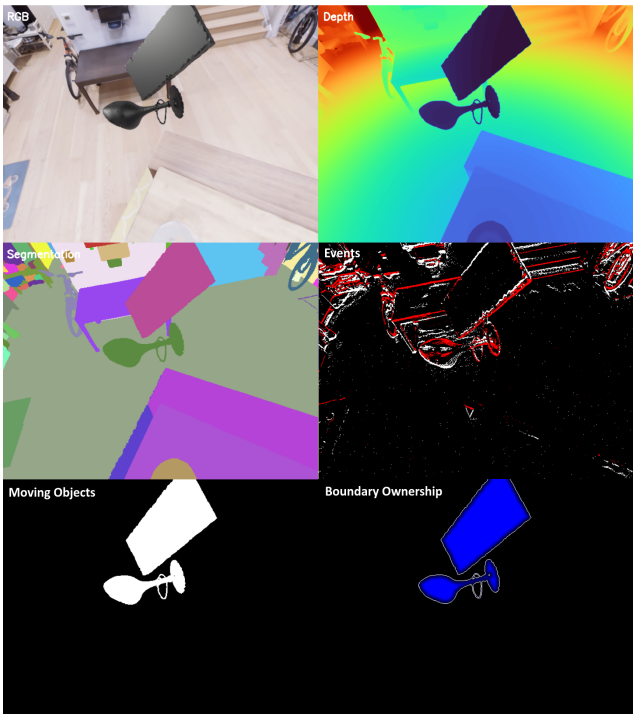


Figure 4. Example synthetic data generated by our Blender pipeline. Each scene provides RGB, depth, instance segmentation, simulated events, moving object masks, and border ownership ground truths—enabling supervision precisely where events occur.

BlenderProc-driven [7] rendering pipeline. Each scene combines photorealistic indoor environments from the Replica dataset [33] with diverse dynamic objects from ShapeNet [6], animated along smooth trajectories to produce natural camera and object motion.

Scene Composition. Each sequence randomly selects a Replica room (apartments, offices, or hotel rooms) that provides realistic static geometry—walls, floors, furniture, and architectural detail. We populate scenes with 3–8 ShapeNet objects, which move along randomized 3D Lissajous trajectories in complex motions. The camera trajectories are generated using orbital paths with randomized radius, height, and look-at targets, augmented with position and orientation jitter to prevent overfitting to synthetic motion patterns. For both camera and object motion we use cubic spline smoothing for physically plausible velocity profiles.

Ground Truth Modalities. We render 640×480 sequences at 1200 real-world-equivalent FPS and simulate events with a log-intensity threshold model (positive/negative thresholds sampled from 0.1–0.3) following V2E [13]. Leak and shot noise are set to 0.5 Hz and 1 Hz. For each frame we generate: RGB images, depth maps, instance-level segmentation, and binary moving-object masks. Instance segmentation supplies precise object boundaries, while depth discontinuities encode occlusions. Combined, these signals allow automatic construction of signed-distance ownership fields that focus supervision on motion boundaries.

Border Ownership Supervision. Synthetic data provides perfect border-ownership ground truth—unavailable in real event datasets due to noise, limited temporal resolution, and calibration errors. At each motion boundary, we know (a) which pixels belong to each object (from instance masks), (b) the exact 3D depth discontinuity (from depth maps), and (c) a high-frequency motion mask. This yields a continuous, differentiable ownership signal precisely at the locations where event cameras deliver measurements.

Dataset Scale. We generate 716 diverse sequences for training, each containing 360-500 frames. The high temporal resolution (1200 FPS) ensures smooth event generation without motion aliasing. Replica's 18 distinct room layouts provide geometric diversity, while ShapeNetv2's 52,472 unique objects prevent overfitting to specific shapes. All scenes are rendered offline with ray-traced lighting, ensuring photorealism that narrows the sim-to-real gap.

5. Evaluation

Evaluation datasets Table 1 compares our synthetic dataset with the real benchmarks. To assess zero-shot generalization, we evaluate the trained model on four real event datasets without fine-tuning:

EVIMO2 [4] provides motion-capture ground truth for motion segmentation, with occasional annotation gaps when tracking is lost.

EVIMO1 [20] is an earlier motion-capture dataset;

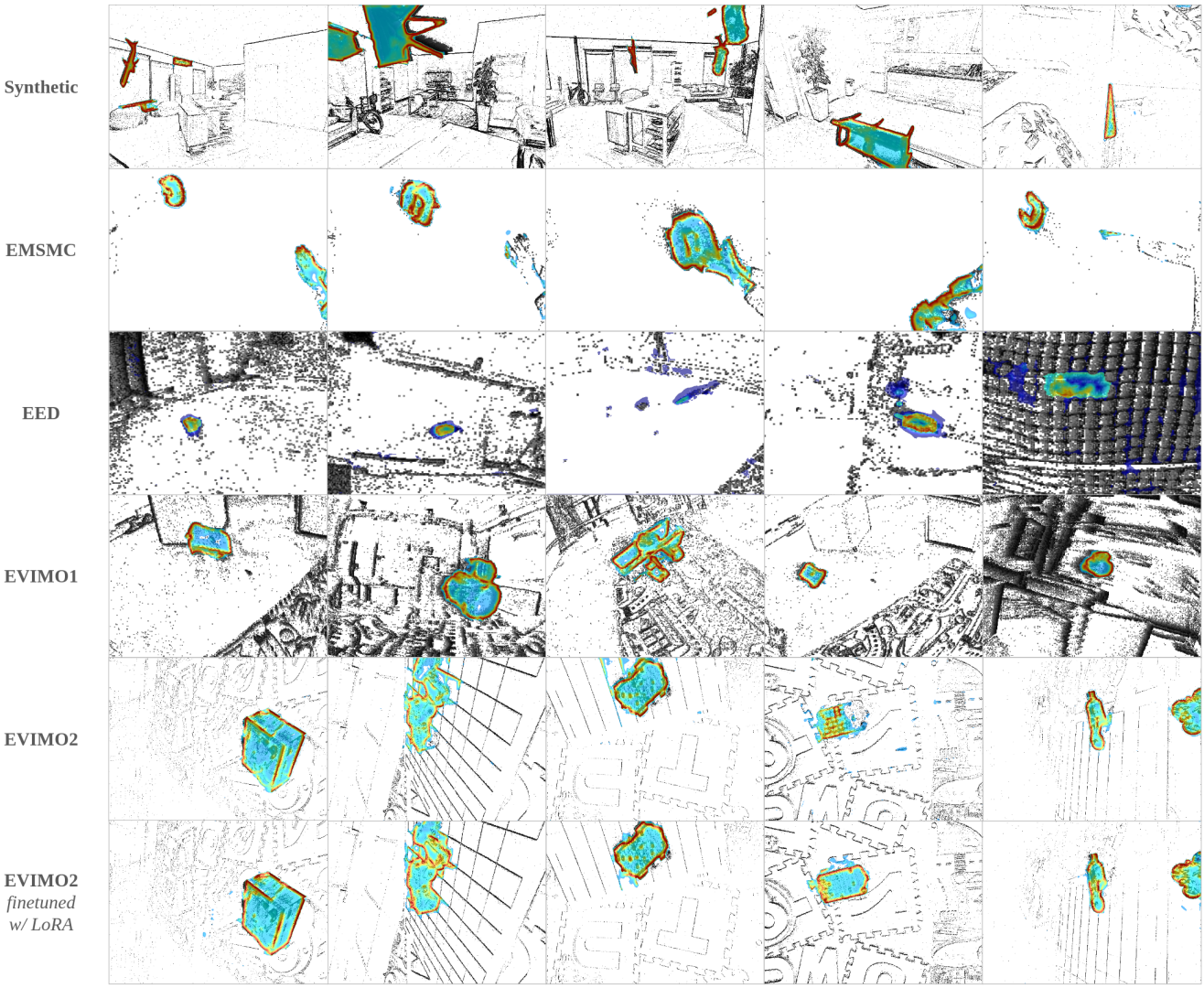


Figure 5. Our synthetically trained model generalizes border and ownership across datasets using the same model weights. We also show the LoRA fine-tuned result on EVIMO2.

mask-event misalignment in some frames prevents reliable border-ownership labels.

EED [19] contains fast-motion sequences under challenging illumination and provides bounding-box annotations for moving objects.

EMSMC [41] contains deformable human motion sequences but does not provide annotations.

5.1. Evaluation Metrics

Intersection over Union (IoU). We evaluate our method using mIoU, following the standard protocol [11, 21, 26, 37]. mIoU is computed as the ratio of intersection over union between predicted and ground truth events, evaluated on frames containing moving objects.

Our network predicts both border ownership and motion masks, supervised by ground truth labels (visualized in Figure 4). We average the border ownership and motion mask (50% each) and apply a fixed threshold to generate binary segmentation masks for mIoU evaluation.

Detection rate. We report detection rates on the EED dataset [19] in Table 3, following the evaluation protocol defined by the dataset authors. Bounding boxes are extracted by fitting to the binary segmentation masks.

6. Results

Setup. Unless otherwise noted, we evaluate the streaming model trained *only* on synthetic events (Sec. 4) using the end-of-window alignment protocol and 10ms time-surface input described in Sec. 3. We report zero-shot transfer on EED, EVIMO1, EVIMO2, and EMSMC, and optional few-shot adaptation on EVIMO2 via ConvLoRA (Sec. 8.1).

Method	Modality	Table	Box	Floor	Plain Wall	Fast Motion	Average↑
MODEL-BASED							
EMSGC [41] Top 30%	Event	55	24	18	24	43	32.8
LEARNING-BASED							
Un-EVIMO [37]	Event+Flow+Depth	50	45	56	53	44	49.6
SpikeMS [26]	Event	50	65	53	63	38	53.8
GConv [21]	Event	51	60	55	**80**	39	57
EVDodgeNet [27]	Event+Flow	70	67	61	72	60	66
EVIMO [20]	Event+Flow+Depth	**79**	70	59	78	**67**	**70.6**
Ours (zero-shot)	Event	74	**77**	**69**	65	63	69.6

Table 2. Performance across scenarios on EVIMO1. Our synthetically-trained model is zero-shot evaluated on the EVIMO1.

Method	Fast Drone	Lighting Var.	Occlusions	Background	Multiple	Avg.↑
SOFAS [31]	88.89	0	80	22.08	46.15	47.74
EED [19]	92.78	84.52	90.83	89.21	87.32	89.34
EMSMC [32]	**96.3**	80.51	92.31	**100**	**96.77**	92.31
EMSGC [41]	**96.3**	**93.51**	**100**	**100**	N/A	N/A
0-MMS [25]	N/A	N/A	N/A	N/A	N/A	**94.2**
Ours (zero-shot)	92.5	72.7	**100**	**100**	84.8	90

Table 3. Detection rates on EED dataset

Method	Modality	mIoU↑
MODEL-BASED		
EMSMC [32]	Event	16.99
EMSGC [41]	Event	64.38
MSEE [14]	Event	77.4
VCM [39]	Event	84
LEARNING-BASED		
UDA [18]	RGB + Event	63.4
SemanticAided [15]	Semantic + Depth	79.82
Ours (zero-shot)	Event	81.55
Ours (fine-tuned)	Event	**85.12**

Table 4. Motion Segmentation on EVIMO2.

6.1. Comparison Methods

We compare against both model-based and learning-based event motion-segmentation methods. Model-based approaches estimate motion parameters by motion compensation or flow-based formulations and then group events into motion layers, while learning-based methods predict motion masks directly from event representations, sometimes with self-labeling or domain adaptation. Tables 2, 3, and 4 report the relevant baselines for each benchmark.

6.2. EVIMO1

Table 2 compares our synthetic-only, event-only model to geometric and learning baselines across the five EVIMO1 scenarios. Our approach, without training on EVIMO1, attains a **69.6** average mIoU, competitive with the EVIMO pipeline that leverages event+flow+depth (**70.6**) while exceeding prior event-only learners (e.g., SpikeMS: 53.8, GConv: 57, EVDodgeNet: 66). Per-scenario results show strong performance on *Table*, *Box*, and *Floor*, with smaller gaps on *Plain Wall* and *Fast Motion*. These gains indicate that explicitly learning ownership along motion boundaries confers much of the robustness traditionally attributed to geometry, without auxiliary modalities.

6.3. EED

On EED (Table 3), our zero-shot detector achieves an average detection rate of **90%**, with perfect scores on *Occlusions* and *Background*; and strong performance on *Fast Drone*. Performance under *Lighting Variation* is comparatively lower, consistent with the sensitivity of event generation to illumination changes. This highlights a concrete failure mode under illumination-driven event-statistics shift, which we discuss further in Sec. 8. The pattern supports our premise: ownership-aware border reasoning is particularly advantageous for scenes with inter-object occlusions and complex figure–ground relationships.

6.4. EVIMO2

Table 4 reports mIoU on EVIMO2. *Zero-shot*, our event-only model reaches **81.55** mIoU—surpassing several learning baselines that depend on additional cues (e.g., SemanticAided using semantics+depth: 79.82; UDA using RGB+Event: 63.4) and nearing the best model-based result (VCM: 84). With *few-shot* ConvLoRA adaptation, mIoU rises to **85.12**, exceeding VCM while keeping the backbone frozen and updating only a small fraction of parameters. Figure 5 illustrates the qualitative improvements: sharper owned boundaries and cleaner motion blobs across diverse scenes and sensors, including the LoRA-adapted EVIMO2 sequences.

6.5. EMSMC and Qualitative Analysis

The EMSMC dataset does not provide annotations; qualitative analysis shown in Figure 5 visualizes zero-shot overlays

on EMSMC, EED, and EVIMO1/2. With LoRA finetuning on EVIMO2, the boundary becomes even sharper and more complete. The model localizes moving-object edges and assigns ownership consistently across different resolutions, motion regimes (handheld, drone, sliding track), and sensor families.

6.6. Runtime and robustness to event rates

	0-MMS	EMSMC	EV-IMO	EMSGC	Ours
events/ms	1,160	3,963	5,410	5	6,522

Table 5. Event computation throughput across methods.

Consistent with Sec. 3.4, our 16.8M-parameter model streams in real time in our implementation. Using 10 ms event slices in all experiments, we achieve an effective throughput of 6,522 events/ms on EVIMO2, which is recorded with a high-resolution event sensor (Table 5). On an NVIDIA RTX2080Ti, this corresponds to an inference rate of 200 FPS, i.e., roughly 2× faster than real time.

7. Ablation Study

7.1. Is border-ownership supervision more effective than only a binary mask?

Setup. We compare training our network with full border-ownership supervision against training with only a binary motion mask, keeping the architecture and training data fixed.

Results. Our experimental evaluation found that without border-ownership supervision, the model attains 76.00 mIoU on EVIMO2 in the zero-shot setting, substantially below the 81.55 mIoU achieved when border ownership is included (Table 4). Additionally, we observe less precise borders and more false positive noise in the binary motion mask-only predictions.

Takeaway. Incorporating border-ownership supervision supplies substantially richer spatial cues than a binary motion mask alone, yielding a 5.55 mIoU performance improvement under domain transfer.

7.2. Does synthetic training help?

Setup. Train the same network on only EVIMO2 (real) vs. train only on scenes rendered synthetically; evaluate both on the EVIMO2 test split.

Results. Real-only training achieves 74.52 mIoU, whereas synthetic-only training reaches 81.55 mIoU (shown in Table 6).

Takeaway. Even with EVIMO2 ground truth available, synthetic training is more effective: it provides cleaner border/ownership supervision and much larger, noise-free coverage, yielding smoother gradients and higher accuracy.

7.3. Is frame-based rendering a bottleneck for event learning?

	EVIMO2	Synthetic	Synthetic-4x	Synthetic-8x
events/ms	3,261	1,878	3,564	3,756
mIoU	74.52	80.02	81.55	81.39

Table 6. Simulated event density vs. prediction accuracy.

Setup. With time-surface inputs and the same synthetic scenes, we use the NN-based frame-interpolation Super-SloMo [16] to upsample the frame rate before event simulation: 1200 FPS, 4800 FPS (4×), and 9600 FPS (8×). We train on each synthetic variant and evaluate zero-shot on EVIMO2.

Results. The event density rises modestly (1,878 → 3,564 → 3,756 events/ms), while mIoU changes are marginal (80.02 → 81.55 → 81.39; Table 6).

Takeaway. With a time-surface representation and short windows, increasing the frame-based render rate yields diminishing returns: once the per-frame pixel motion is ≤1 px, extra frames do not create meaningfully richer events. We observe no practical bottleneck from frame-based rendering under this regime.

8. Discussion and Conclusion

Discussion and limitations. Event cameras provide the strongest evidence at motion boundaries, which is why border-ownership supervision improves dense motion segmentation. At the same time, this makes the method most reliable when moving-object boundaries generate clear and stable event structure. Additional zero-shot analysis indicates that degradation is more pronounced in slow-motion and high-noise cases, whereas fast motion is less problematic for the current representation. Performance can degrade when this assumption weakens, including slow motion, high sensor noise, challenging illumination, and motion-parallax configurations in which motion cues alone are insufficient to disambiguate ownership. More generally, the current formulation is designed for ownership-aware segmentation of *moving* objects, rather than a general treatment of all static figure–ground boundaries under pure ego-motion. Finally, although ConvLoRA enables effective few-shot adaptation, we validate it only on EVIMO2; broader adaptation across sensors and environments remains future work.

Future work. Future work will extend ownership-aware supervision to additional event-based tasks, including optical flow, tracking, and structure-from-motion, and study stronger adaptation strategies across sensors and scenes.

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